

RequestIdFilter.java

```
package com.harisudhan.harisudhanmart.filter;

import java.io.IOException;
import java.util.UUID;
import javax.servlet.Filter;
import javax.servlet.FilterChain;
import javax.servlet.FilterConfig;
import javax.servlet.ServletException;
import javax.servlet.ServletRequest;
import javax.servlet.ServletResponse;
import javax.servlet.annotation.WebFilter;
import org.slf4j.MDC;

/** Generates a per-request correlation id and attaches it to the SLF4J MDC (Section 18). */
@WebFilter("/*")
public class RequestIdFilter implements Filter {

    private static final String MDC_KEY = "requestId";

    @Override
    public void init(FilterConfig filterConfig) { }

    @Override
    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
        throws IOException, ServletException {
        String requestId = UUID.randomUUID().toString();
        MDC.put(MDC_KEY, requestId);
        try {
            chain.doFilter(request, response);
        } finally {
            MDC.remove(MDC_KEY);
        }
    }

    @Override
    public void destroy() { }
}
```